

MEMEHUB: A SOCIAL MEDIA PLATFORM FOR MEME SHARING

PROJECT REPORT

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ABSTRACT

MemeHub is a full-stack social media web application developed to provide users with a platform for sharing memes, interacting with other users, and building social connections. The platform enables users to create accounts, upload memes, like posts, comment on content, follow other users, and manage personal profiles.

The application is built using the MERN stack, consisting of MongoDB for database management, Express.js and Node.js for backend development, and React.js for frontend development. Cloudinary is integrated for image storage and management, while JSON Web Tokens (JWT) are used for secure authentication and authorization.

The project demonstrates modern web development practices, RESTful API design, user authentication, database relationships, and cloud-based media storage.

INTRODUCTION

Social media platforms have become a significant part of digital communication. Users share content, interact with communities, and connect with people worldwide. MemeHub is designed as a simplified social networking platform focusing on meme sharing and social interaction.

The primary objective of the project is to provide a user-friendly environment where users can create profiles, share content, and engage with other users through likes, comments, and follow systems.

OBJECTIVES

The main objectives of the project are:

1. To develop a full-stack social media platform.
 2. To implement secure user authentication.
 3. To allow users to upload and manage meme content.
 4. To facilitate social interaction through likes and comments.
 5. To implement a follow/unfollow mechanism.
 6. To provide personalized user profiles.
 7. To deploy the application on cloud platforms.
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TECHNOLOGIES USED

Frontend

- React.js
- React Router DOM
- Axios
- CSS

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas

Authentication

- JSON Web Token (JWT)

Cloud Services

- Cloudinary (Image Storage)
 - Render (Backend Hosting)
 - Vercel (Frontend Hosting)
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SYSTEM ARCHITECTURE

The system follows a client-server architecture.

Client Side

- React frontend
- User interface and interaction handling
- API communication using Axios

Server Side

- Express.js REST API
- Authentication and authorization
- Business logic implementation

Database Layer

- MongoDB Atlas
- Stores users, memes, comments, and follower relationships

Cloud Storage

- Cloudinary stores uploaded images and profile pictures
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MODULES OF THE SYSTEM

User Authentication Module

Features:

- User Registration
- User Login
- JWT Authentication
- Protected Routes

Profile Management Module

Features:

- View User Profile
- Upload Profile Picture
- Display Followers and Following Count
- View User Posts

Meme Management Module

Features:

- Upload Meme
- View Meme Feed
- Delete Own Meme
- View Other Users' Memes

Social Interaction Module

Features:

- Like Memes
 - Comment on Memes
 - Delete Comments
 - Follow Users
 - Unfollow Users
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DATABASE DESIGN

User Collection

Attributes:

- _id
- username
- email
- password
- avatar
- bio
- followers
- following
- createdAt
- updatedAt

Meme Collection

Attributes:

- _id
- title

- imageUrl
- user
- likes
- commentCount
- createdAt

Comment Collection

Attributes:

- _id
 - text
 - user
 - meme
 - createdAt
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IMPLEMENTATION DETAILS

Authentication

JWT tokens are generated upon successful login and stored on the client side. Protected routes verify the authenticity of the user before granting access.

Image Upload

Users upload memes and profile pictures through Cloudinary integration. Uploaded images are stored securely in cloud storage, and image URLs are saved in MongoDB.

Follow System

The follow/unfollow mechanism updates follower and following arrays in the User collection, enabling social networking functionality.

Comments and Likes

Users can engage with content through likes and comments. The platform dynamically updates interaction counts and displays associated user information.

FEATURES IMPLEMENTED

- User Registration
- User Login

- JWT Authentication
 - Meme Upload
 - Meme Feed
 - Like System
 - Comment System
 - Delete Comments
 - Delete Own Memes
 - User Profiles
 - Profile Pictures
 - Follow/Unfollow System
 - Profile Navigation
 - Cloud Image Storage
 - Full Deployment
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TESTING

The application was tested for the following functionalities:

Test Case	Result
User Registration	Passed
User Login	Passed
Meme Upload	Passed
Meme Deletion	Passed
Like Feature	Passed
Comment Feature	Passed
Follow User	Passed
Unfollow User	Passed
Profile Visit	Passed
Avatar Upload	Passed

ADVANTAGES

- User-friendly interface
- Secure authentication system
- Cloud-based image storage
- Responsive design

- Scalable architecture
 - Real-time social interaction simulation
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LIMITATIONS

- No direct messaging system
 - No notification feature
 - No user search functionality
 - No real-time updates
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FUTURE ENHANCEMENTS

1. Real-time Messaging
 2. User Search System
 3. Notification System
 4. Infinite Scrolling Feed
 5. Edit Profile Functionality
 6. Dark/Light Theme Support
 7. Real-Time Chat Using Socket.IO
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CONCLUSION

MemeHub successfully demonstrates the implementation of a modern social media platform using the MERN stack. The project incorporates essential social networking features such as user authentication, content sharing, social interaction, and profile management. Through the integration of cloud services and deployment platforms, the application provides a complete full-stack web development solution suitable for real-world use cases and academic learning.

REFERENCES

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